

GENDER BIAS & FAIRNESS IN ML- BASED LOAN APPROVAL

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PROJECT OVERVIEW

Dataset: 45,000 synthetic loan applications (44.8% female, 55.2% male)

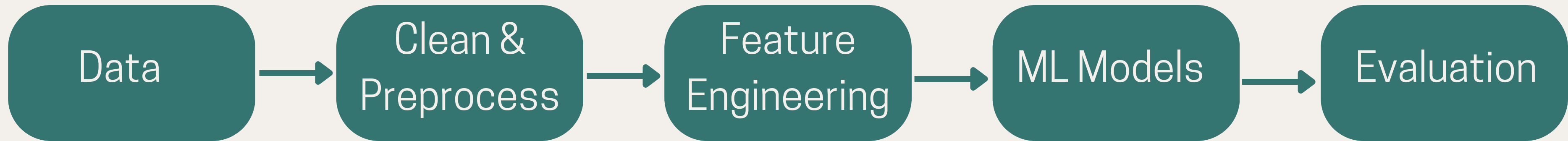
Problem: Binary classification, predicting loan status

Imbalance: 77.78% rejected, 22.22% approved

OBJECTIVES

- A** Build a reproducible ML pipeline in Python
- B** Apply domain-aware feature engineering
- C** Benchmark 4 classifier
- D** Evaluate fairness via DPD, DIR, EOD metrics

SYSTEM DIAGRAM & MODELS



Models Used:



MAIN RESULT:

GRADIENT BOOSTING WAS THE BEST FINAL MODEL AFTER TUNING.

TUNED MODELS RESULTS:

Model	Accuracy	F1-score	ROC-AUC
Logistic Regression	0.9031	0.7781	0.9575
Decision Tree	0.9198	0.8038	0.9573
Random Forest	0.9300	0.8309	0.9756
Gradient Boosting	0.9366	0.8477	0.9787

- Gradient Boosting achieved the highest Accuracy, F1-score, and ROC-AUC.
- Random Forest was the second-best model and remained very competitive.
- Class 1 was harder to predict because the dataset is imbalanced.
- Fairness results showed very small gender disparity: Demographic Parity Difference and Equal Opportunity Difference close to 0, Disparate Impact Ratio close to 1.
- Logistic Regression had the most balanced fairness profile, but Gradient Boosting had the best predictive performance.



THANK YOU!